# Master's Thesis Notes

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# 1 Formalization of the Neural Tangent Kernel (NTK)

## 1.1 Setup and Assumptions

We consider a supervised learning setting with data $\{(x_i, y_i)\}_{i=1}^n$, where $x_i \in \mathbb{R}^d$ and $y_i \in \mathbb{R}$. A neural network $f : \mathbb{R}^d \times \Theta \to \mathbb{R}$ is parameterized by $\theta \in \Theta \subseteq \mathbb{R}^p$, where $p$ is the number of parameters.

**Assumption 1.1** (Model and training)**.**

1. (*Differentiability*) The network $f(x;\theta)$ is differentiable in $\theta$.
2. (*Random initialization*) We initialize parameters $\theta_0$ i.i.d. with zero mean and variance scaled according to layer input dimensions (e.g. NTK schemes), so that activations and gradients remain well-behaved as depth/width grow [3, 1, 2].
3. (*Training*) We train $f$ by gradient descent on the squared loss:

$$L(\theta) = \frac{1}{2} \sum_{i=1}^{n} \big(f(x_i;\theta) - y_i\big)^2.$$

### 1.2 Linearization around Initialization

A first-order Taylor expansion of $f(x;\theta)$ around $\theta_0$ gives

$$f(x;\theta) \;\approx\; f(x;\theta_0) + \nabla_\theta f(x;\theta_0)^\top(\theta-\theta_0).$$

- $f(x;\theta_0)$ is the network output at initialization (a bias term).
- $\phi(x) := \nabla_\theta f(x;\theta_0)$ is the feature vector induced at initialization.

Thus, locally, the network behaves as a linear model in $\theta$:

$$f(x;\theta) \approx f(x;\theta_0) + \phi(x)^\top(\theta-\theta_0). \tag{1.1}$$

### 1.3 Neural Tangent Kernel

**Definition 1.2** (Neural Tangent Kernel [1])**.** Given initialization $\theta_0$, the Neural Tangent Kernel (NTK) is

$$K(x,x') \;=\; \nabla_\theta f(x;\theta_0)^\top \nabla_\theta f(x';\theta_0).$$

The NTK captures how parameter updates couple the outputs of $x$ and $x'$.

*Remark* 1.3.

- $K(x,x')$ is positive semidefinite [1].
- In the infinite-width limit, under common initializations, $K(x,x')$ converges almost surely to a deterministic kernel depending only on architecture and activation [1, 2].
- For finite but large width, $K$ is still a random kernel due to random initialization, but it concentrates around its infinite-width expectation. Fluctuations vanish at rate $O(1/\sqrt{m})$ as width $m\to\infty$ [1, 2].

### 1.4 NTK Characterization (one hidden layer, no biases)

Consider a width-$m$ one-hidden-layer network

$$f(x) \;=\; \frac{1}{\sqrt{m}}\sum_{r=1}^{m} a_r\,\sigma(w_r^\top x),$$

with parameters $\theta = \{(a_r,w_r)\}_{r=1}^m$, activation $\sigma$, and random initialization

$$a_r \sim \mathcal{N}(0,\sigma_a^2), \qquad w_r \sim \mathcal{N}\Big(0,\frac{\sigma_w^2}{d}I_d\Big),$$

independently across $r$.

#### Finite-width NTK at initialization

By definition,

$$K_m(x,x') \;=\; \nabla_\theta f(x)^\top \nabla_\theta f(x') \;=\; \sum_{r=1}^{m}\left[\underbrace{\nabla_{a_r} f(x)\,\nabla_{a_r} f(x')}_{\text{output-weight part}} + \underbrace{\nabla_{w_r} f(x)^\top \nabla_{w_r} f(x')}_{\text{hidden-weight part}}\right].$$

Compute the gradients:

$$\nabla_{a_r} f(x) = \frac{1}{\sqrt{m}} \sigma(w_r^\top x), \qquad \nabla_{w_r} f(x) = \frac{1}{\sqrt{m}}\, a_r\, \sigma'(w_r^\top x)\, x.$$

Hence,

$$\boxed{K_m(x,x') = \frac{1}{m}\sum_{r=1}^{m} \sigma(w_r^\top x)\, \sigma(w_r^\top x') + \frac{1}{m}\sum_{r=1}^{m} a_r^2\, \sigma'(w_r^\top x)\, \sigma'(w_r^\top x')\, x^\top x'}.$$

**Infinite-width limit**

The two sums are empirical averages of i.i.d. terms. Since $a_r$ and $w_r$ are independent with finite moments, the (strong) law of large numbers gives, almost surely,

$$\frac{1}{m}\sum_{r=1}^{m} \sigma(w_r^\top x)\, \sigma(w_r^\top x') \;\longrightarrow\; \mathbb{E}_w\big[\sigma(w^\top x)\, \sigma(w^\top x')\big],$$

$$\frac{1}{m}\sum_{r=1}^{m} a_r^2\, \sigma'(w_r^\top x)\, \sigma'(w_r^\top x') \;\longrightarrow\; \sigma_a^2\, \mathbb{E}_w\big[\sigma'(w^\top x)\, \sigma'(w^\top x')\big].$$

Thus, in the infinite-width limit, the empirical NTK converges almost surely to a deterministic kernel [1, 2].

$$\boxed{K_\infty(x,x') = \mathbb{E}_w\big[\sigma(w^\top x)\, \sigma(w^\top x')\big] \;+\; \sigma_a^2\, x^\top x'\, \mathbb{E}_w\big[\sigma'(w^\top x)\, \sigma'(w^\top x')\big]}$$

with $w \sim \mathcal{N}(0, \frac{\sigma_w^2}{d} I_d)$.

## 1.5 One Hidden Layer: Adding Biases

If we allow per-neuron biases, we get minimal changes from subsection 1.4. Consider

$$f(x) \;=\; \frac{1}{\sqrt{m}}\sum_{r=1}^{m} a_r\, \sigma\big(w_r^\top x + b_r\big), \quad a_r \sim \mathcal{N}(0,\sigma_a^2),\; w_r \sim \mathcal{N}\Big(0, \tfrac{\sigma_w^2}{d} I_d\Big),\; b_r \sim \mathcal{N}(0,\sigma_b^2),$$

independently across $r$. Let $u_r(x) := w_r^\top x + b_r$.

**Finite width** Gradients are

$$\nabla_{a_r} f(x) = \tfrac{1}{\sqrt{m}} \sigma(u_r(x)), \quad \nabla_{w_r} f(x) = \tfrac{1}{\sqrt{m}}\, a_r\, \sigma'(u_r(x))\, x, \quad \nabla_{b_r} f(x) = \tfrac{1}{\sqrt{m}}\, a_r\, \sigma'(u_r(x)).$$

Thus

$$\boxed{K_m(x,x') = \frac{1}{m}\sum_{r=1}^{m} \sigma(u_r(x))\, \sigma(u_r(x')) + \frac{1}{m}\sum_{r=1}^{m} a_r^2\, \sigma'(u_r(x))\, \sigma'(u_r(x'))\, \big(x^\top x' + 1\big)}.$$

**Infinite width (preactivation covariance picks up $\sigma_b^2$).** With $(U,V)$ jointly Gaussian:

$$\mathrm{Var}(U) = \tfrac{\sigma_w^2}{d}\|x\|^2 + \sigma_b^2, \quad \mathrm{Var}(V) = \tfrac{\sigma_w^2}{d}\|x'\|^2 + \sigma_b^2, \quad \mathrm{Cov}(U,V) = \tfrac{\sigma_w^2}{d} x^\top x' + \sigma_b^2,$$

we have

$$\boxed{K_\infty(x,x') = \mathbb{E}\big[\sigma(U)\sigma(V)\big] + \sigma_a^2\, \big(x^\top x' + 1\big)\, \mathbb{E}\big[\sigma'(U)\sigma'(V)\big]}.$$

### 1.6 Extension to Deep Fully–Connected Networks

**No biases.** Let $n_0 = d,\ n_1, \dots, n_{L-1}$ be layer widths and define

$$\alpha^{(0)}(x) = x, \qquad \tilde{\alpha}^{(\ell+1)}(x) = W^{(\ell)}\alpha^{(\ell)}(x), \qquad \alpha^{(\ell+1)}(x) = \sigma\big(\tilde{\alpha}^{(\ell+1)}(x)\big),$$

for $\ell = 0, \dots, L-2$, with entries $W^{(\ell)} \overset{\text{i.i.d.}}{\sim} \mathcal{N}\Big(0, \frac{\sigma_w^2}{n_\ell}\Big)$. The scalar output is

$$f_\theta(x) = \frac{1}{\sqrt{n_{L-1}}} \sum_{r=1}^{n_{L-1}} a_r\, \alpha_r^{(L-1)}(x), \qquad a_r \sim \mathcal{N}(0, \sigma_a^2).$$

Define the activation covariance

$$\Sigma^{(\ell)}(x, x') := \mathbb{E}\big[\alpha_r^{(\ell)}(x)\, \alpha_r^{(\ell)}(x')\big], \qquad q^{(\ell)}(x) := \Sigma^{(\ell)}(x, x),$$

with base $\Sigma^{(0)}(x, x') = \frac{1}{d}\, x^\top x'$. Then the forward (NNGP) recursion is

$$\boxed{\Sigma^{(\ell+1)}(x, x') = \mathbb{E}_{f \sim \mathcal{GP}\left(0,\ \sigma_w^2\, \Sigma^{(\ell)}\right)}\big[\sigma\big(f(x)\big)\, \sigma\big(f(x')\big)\big]}.$$

Define the derivative–correlation kernel

$$\boxed{\dot{\Sigma}^{(\ell+1)}(x, x') := \mathbb{E}_{f \sim \mathcal{GP}\left(0,\ \sigma_w^2\, \Sigma^{(\ell)}\right)}\big[\sigma'\big(f(x)\big)\, \sigma'\big(f(x')\big)\big]}.$$

The limiting NTK satisfies

$$\boxed{\Theta_\infty^{(1)}(x, x') = \Sigma^{(1)}(x, x'), \qquad \Theta_\infty^{(\ell+1)}(x, x') = \Theta_\infty^{(\ell)}(x, x')\, \dot{\Sigma}^{(\ell+1)}(x, x') + \Sigma^{(\ell+1)}(x, x')}.$$

For a dataset $\{x_i\}$:

$$\Theta^{(\ell+1)} = \Theta^{(\ell)} \odot \dot{\Sigma}^{(\ell+1)} + \Sigma^{(\ell+1)}, \qquad \Theta^{(1)} = \Sigma^{(1)}.$$

Setting $L = 2$ recovers the two-term one-layer kernel.

### 1.7 Training Dynamics under NTK

We train with squared loss

$$L(\theta) \;=\; \frac{1}{2} \sum_{i=1}^{n} \big(f(x_i; \theta) - y_i\big)^2,$$

and consider *gradient flow* in parameter space, i.e. the continuous-time limit of gradient descent as the step size $\eta \to 0$:

$$\frac{d\theta_t}{dt} \;=\; -\nabla_\theta L(\theta_t).$$

Let $f_t(x_i) := f(x_i; \theta_t)$. By the chain rule,

$$\frac{d}{dt} f_t(x_i) \;=\; \nabla_\theta f(x_i; \theta_t)^\top \frac{d\theta_t}{dt} \;=\; -\nabla_\theta f(x_i; \theta_t)^\top \nabla_\theta L(\theta_t).$$

Compute the parameter gradient of the loss:

$$\nabla_\theta L(\theta_t) \;=\; \sum_{j=1}^{n} \big(f_t(x_j) - y_j\big)\, \nabla_\theta f(x_j; \theta_t).$$

Substituting gives

$$\frac{d}{dt} f_t(x_i) \;=\; -\sum_{j=1}^{n} \underbrace{\nabla_\theta f(x_i;\theta_t)^\top \nabla_\theta f(x_j;\theta_t)}_{=:\; K_t(x_i,x_j)} \big(f_t(x_j) - y_j\big).$$

Stacking $f_t = (f_t(x_1), \dots, f_t(x_n))$ yields the vector ODE

$$\frac{d}{dt} f_t \;=\; -K_t\,(f_t - y),$$

where $[K_t]_{ij} = K_t(x_i, x_j)$ is the (time–dependent) NTK matrix.

**Constant–kernel (NTK) regime.** In the infinite–width limit (or under a lazy–training approximation), the kernel remains essentially constant during training, $K_t \approx K_0 =: K$ [1]. The ODE reduces to

$$\frac{d}{dt} f_t \;=\; -K\,(f_t - y).$$

Let $r_t := f_t - y$. Then $\dot{r}_t = -K r_t$ with solution $r_t = e^{-Kt} r_0$, i.e.

$$f_t \;=\; y + e^{-Kt}\,(f_0 - y).$$

The convergence rate along eigenvector $v_j$ of $K$ is exponential with rate $\lambda_j$, the corresponding eigenvalue.

### 1.8 Lazy Training Regime

Training is in the *lazy regime* if parameter updates stay small relative to initialization:

$$\|\theta_t - \theta_0\| \ll \|\theta_0\|.$$

Then $\phi(x)$ and the NTK remain essentially constant and training is equivalent to kernel regression with fixed kernel $K$. When $\|\theta_t - \theta_0\|$ is not negligible, $\phi(x)$ evolves, yielding adaptive feature learning beyond NTK.

## 2 Population ReLU Kernel on the Circle

Throughout, $(\cdot|\cdot)$ denotes the Euclidean inner product on $\mathbb{R}^d$.

### 2.1 Population ReLU Feature Kernel

Let $\sigma(t) = \max\{0, t\}$ denote the ReLU activation and let $w \sim \mathcal{N}(0, I_d)$. We define the *population ReLU feature kernel* by

$$K(x, y) \;:=\; \mathbb{E}_w\big[\sigma((w|x))\sigma((w|y))\big], \qquad x, y \in \mathbb{R}^d. \tag{2.1}$$

This kernel is the covariance of the random feature map $\phi_w(x) = \sigma((w|x))$ and is therefore positive semidefinite.

Let $\gamma_d$ denote the standard Gaussian measure on $\mathbb{R}^d$,

$$d\gamma_d(w) = (2\pi)^{-d/2} \exp\Big(-\frac{\|w\|^2}{2}\Big)\, dw.$$

Then (2.1) can be written as the Gaussian integral

$$K(x,y) = \int_{\mathbb{R}^d} \sigma((w|x))\sigma((w|y))\, d\gamma_d(w). \tag{2.2}$$

Using $\sigma(t) = t\,\mathbf{1}_{\{t>0\}}$, this becomes

$$K(x,y) = \int_{\mathbb{R}^d} (w|x)(w|y)\,\mathbf{1}_{\{(w|x)>0\}}\,\mathbf{1}_{\{(w|y)>0\}}\, d\gamma_d(w). \tag{2.3}$$

## 2.2 Reduction to a Two-Dimensional Gaussian

Let $V = \operatorname{span}\{x,y\} \subset \mathbb{R}^d$.

**Lemma 2.1** (Orthogonal decomposition)**.** *For every $w \in \mathbb{R}^d$ there exist unique vectors $w_0 \in V$ and $w_\perp \in V^\perp$ such that*

$$w = w_0 + w_\perp, \qquad (w_0|w_\perp) = 0.$$

*Proof.* Choose an orthonormal basis $\{e_1,\dots,e_k\}$ of $V$ and extend it to an orthonormal basis $\{e_1,\dots,e_d\}$ of $\mathbb{R}^d$. Writing $w = \sum_{i=1}^d (w|e_i)e_i$, define

$$w_0 := \sum_{i=1}^{k} (w|e_i)e_i, \qquad w_\perp := \sum_{i=k+1}^{d} (w|e_i)e_i.$$

Then $w = w_0 + w_\perp$, $(w_0|w_\perp) = 0$, and uniqueness follows from the uniqueness of the orthonormal expansion. □

Since $x, y \in V$, we have $(w_\perp|x) = (w_\perp|y) = 0$, hence

$$(w|x) = (w_0|x), \qquad (w|y) = (w_0|y).$$

Thus the integrand in (2.3) depends only on $w_0$.

**Lemma 2.2** (Gaussian factorization)**.** *If $w \sim \mathcal{N}(0, I_d)$, then $w_0$ and $w_\perp$ are independent Gaussian vectors with*

$$w_0 \sim \mathcal{N}(0, I_V), \qquad w_\perp \sim \mathcal{N}(0, I_{V^\perp}).$$

*Proof.* In an orthonormal basis adapted to $V \oplus V^\perp$, the coordinates of $w$ are independent standard Gaussians. Grouping coordinates yields the claim. □

Hence integration over $V^\perp$ contributes a factor of one, and the kernel reduces to a Gaussian integral over $V$.

## 2.3 Choice of Coordinates and Angle Representation

Assume $x \neq 0$ and define

$$u := \frac{x}{\|x\|}.$$

If $y$ is not collinear with $x$, define

$$v := \frac{y - (y|u)u}{\|y - (y|u)u\|}.$$

Then $\{u, v\}$ is an orthonormal basis of $V$.

Define the angle $\theta \in [0,\pi]$ by

$$\cos\theta := \frac{(x|y)}{\|x\|\|y\|}.$$

**Lemma 2.3** (Angle decomposition)**.** *With $u, v$ as above,*

$$x = \|x\|u, \qquad y = \|y\|(\cos\theta\, u + \sin\theta\, v).$$

*Proof.* Since $y \in V$,

$$y = (y|u)u + (y|v)v.$$

Now $(y|u) = \|y\|\cos\theta$. By Pythagoras,

$$(y|v)^2 = \|y\|^2 - (y|u)^2 = \|y\|^2 \sin^2\theta,$$

and by construction $(y|v) \geq 0$, hence $(y|v) = \|y\|\sin\theta$. □

## 2.4 Gaussian Coordinates in the Plane

**Lemma 2.4** (Coordinates of $w_0$)**.** *There exist unique scalars $Z_1, Z_2$ such that*

$$w_0 = Z_1 u + Z_2 v, \qquad Z_1 = (w|u),\ Z_2 = (w|v).$$

*Proof.* Since $\{u, v\}$ is an orthonormal basis of $V$, write $w_0 = Z_1 u + Z_2 v$. Taking inner products with $u$ and $v$ gives $Z_1 = (w_0|u)$ and $Z_2 = (w_0|v)$. Because $w = w_0 + w_\perp$ and $u, v \perp w_\perp$, these equal $(w|u)$ and $(w|v)$. □

**Lemma 2.5** (Distribution of $(Z_1, Z_2)$)**.** *Let $w \sim \mathcal{N}(0, I_d)$. Then*

$$(Z_1, Z_2) \sim \mathcal{N}(0, I_2),$$

*and in particular $Z_1, Z_2$ are independent standard normal random variables.*

*Proof.* For any $a \in \mathbb{R}^d$, $(a|w) \sim \mathcal{N}(0, \|a\|^2)$. Thus $Z_1, Z_2 \sim \mathcal{N}(0, 1)$. Moreover,

$$\mathrm{Cov}(Z_1, Z_2) = \mathbb{E}[(w|u)(w|v)] = (u|v) = 0.$$

Since any linear combination $aZ_1 + bZ_2 = (w|au + bv)$ is Gaussian, the vector $(Z_1, Z_2)$ is jointly Gaussian. Zero covariance implies independence. □

Combining the above,

$$(w|x) = \|x\|Z_1, \qquad (w|y) = \|y\|(\cos\theta\, Z_1 + \sin\theta\, Z_2).$$

## 2.5 Evaluation of the Gaussian Integral

Substituting into (2.3) yields

$$K(x, y) = \|x\|\|y\|\,\mathbb{E}\Big[Z_1(\cos\theta\, Z_1 + \sin\theta\, Z_2)\mathbf{1}_{\{Z_1>0\}}\mathbf{1}_{\{\cos\theta Z_1 + \sin\theta Z_2 > 0\}}\Big], \tag{2.4}$$

where $(Z_1, Z_2) \sim \mathcal{N}(0, I_2)$.

The joint density of $(Z_1, Z_2)$ is

$$\frac{1}{2\pi}\exp\Big(-\frac{Z_1^2 + Z_2^2}{2}\Big).$$

Introduce polar coordinates $(Z_1, Z_2) = (R\cos\varphi, R\sin\varphi)$ with $R \geq 0$ and $\varphi \in [0, 2\pi)$. Then the Gaussian measure becomes

$$\frac{1}{2\pi} e^{-R^2/2} R\, dR\, d\varphi,$$

and

$$\cos\theta\, Z_1 + \sin\theta\, Z_2 = R\cos(\varphi - \theta).$$

The indicator constraints are equivalent to

$$\cos\varphi > 0, \qquad \cos(\varphi - \theta) > 0,$$

which define the angular interval

$$\varphi \in (0, \pi) \cap (\theta, \theta + \pi),$$

of length $\pi - \theta$.

Separating radial and angular variables gives

$$\mathbb{E}[\cdots] = \frac{1}{2\pi} \int_0^\infty R^3 e^{-R^2/2}\, dR \int_{\mathcal{I}_\theta} \cos\varphi \cos(\varphi - \theta)\, d\varphi,$$

where $\mathcal{I}_\theta$ denotes the above interval. The radial integral evaluates to

$$\int_0^\infty R^3 e^{-R^2/2}\, dR = 2,$$

and the angular integral yields

$$\int_{\mathcal{I}_\theta} \cos\varphi \cos(\varphi - \theta)\, d\varphi = \frac{1}{2}\big(\sin\theta + (\pi - \theta)\cos\theta\big).$$

**Theorem 2.6** (Arc-cosine kernel)**.** *For all $x, y \in \mathbb{R}^d$,*

$$K(x, y) = \frac{\|x\|\|y\|}{2\pi}\big(\sin\theta + (\pi - \theta)\cos\theta\big), \qquad \theta = \arccos\Big(\frac{(x|y)}{\|x\|\|y\|}\Big).$$

## 2.6 Restriction to the Unit Circle

Let $x(\phi) = (\cos\phi, \sin\phi) \in S^1$. Then $(x(\phi)|x(\psi)) = \cos(\phi - \psi)$ and

$$K(x(\phi), x(\psi)) = k(\theta(\phi - \psi)), \qquad k(\theta) = \frac{1}{2\pi}\big(\sin\theta + (\pi - \theta)\cos\theta\big).$$

## 2.7 Population Integral Operator

Let $\mu$ be the uniform probability measure on $[0, 2\pi)$. Define

$$(Tf)(\phi) = \int_0^{2\pi} K(x(\phi), x(\psi)) f(\psi)\, d\mu(\psi).$$

Then $T$ is a circular convolution operator.

### 2.8 Discretization of the Population

Let $\phi_1, \ldots, \phi_n \overset{iid}{\sim} \mathrm{Unif}[0, 2\pi)$ and define

$$(K_n)_{ij} = K(x(\phi_i), x(\phi_j)).$$

Equivalently,

$$(T_n f)_i = \frac{1}{n} \sum_{j=1}^{n} K(x(\phi_i), x(\phi_j)) f(\phi_j).$$

This is a Monte Carlo discretization of the population operator $T$, isolating finite-sample effects while remaining in the infinite-population (kernel) regime.

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